Solution Requirements

Date	15 March 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).
- Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	No authentication required; open-access web application for farmers. Future scope includes user login via email/OTP for personalized history.	Low
2	Authorization levels	Single user role (Farmer/Guest) with full access to prediction features. Admin role planned for future model management.	Low

3	External interfaces	Kaggle dataset API for data sourcing; Scikit-learn for ML model training; Flask REST API for prediction endpoint.	High
4	Transaction processing	User submits soil/environmental parameters via HTTP POST; backend processes input through the trained ML model and returns crop prediction in real-time.	High
5	Reporting & History	Display predicted crop recommendation on the results page. Users can view and manage past predictions via the Previous Crops page.	High
6	Business rules	Input values must fall within valid agricultural ranges (e.g., N: 0–140, P: 5–145, K: 5–205, temp: 8–44°C, humidity: 14–100%, pH: 3.5–10, rainfall: 20–300mm). Invalid values return error messages.	High
7	Compliance	No personal data collected; system complies with general data privacy principles. Dataset is publicly available (Kaggle open license).	Low
8	Other	Application must work offline once deployed locally. Model retraining capability for updated datasets in future versions.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Prediction results must be generated instantly after form submission.	Page loads in < 2 seconds; prediction response < 1 second
2	Scalability	System should handle multiple concurrent users accessing the web app.	Supports up to 100 concurrent users on local Flask server
3	Security & Data Privacy	No sensitive personal data collected; input values are not stored.	No database storage of user inputs; stateless API
4	Reliability & Availability	Application should run without crashes during demonstration.	99% uptime during local deployment
5	Usability & Accessibility	Simple, intuitive web interface accessible to farmers with minimal tech literacy.	Maximum 3-click workflow from landing to prediction
6	Other (Maintainability)	Code should be modular and well-documented for future enhancements.	Model can be retrained and replaced without code changes